C programming

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Union

- Union is user defined data-type.
- Like struct it is collection of similar or non-similar data elements.
- All members of union share same memory space i.e. modification of an member can affect others too.
- Size of union = Size of largest element
- When union is initialized at declaration, the first member is initialized.
- Application:
  - System programming: to simulate register sharing in the hardware.
  - Application programming: to use single member of union as per requirement.

```
union test
{
    int num;
    char arr[2];
}u = { 65 };

printf("%d, %c, %s\n", u.num, u.arr[0], u.arr);
```

Union and Structure

- Unions and structures can be nested within each other.

```
struct
{
        short s[5];
        union
        {
        char x;
        float y;
        long z;
        short int z1;
        }u;
}t;
```

Function Pointer:

- Function Pointer is **pointers to the function**.
- a function pointer points to code, not data. Typically a function pointer stores the start of executable code.

  we do not allocate de-allocate memory using function pointers.
- A function’s name can also be used to get functions’ address.
- A function can also be passed as an arguments and can be returned from a function.
- Like normal pointers, we can have an array of function pointers.

Function Pointer:

```
#include<stdio.h>

void display(int num1)
{
    printf("\n value of num1=%d",num1);
}

int main()
{
    void (*ptr)(int)=&display;//declaring the function pointer
    //or
  //  void (*ptr)(int)=display;   ;//declaring the function pointer
    (*ptr)(100);//calling the function with function mointer
}
```

pointer :- variable to store/hold an address another var(data)

function pointer:- pointer var which will store an address of a function(code)

address of function
1. name of function
2. & name of function

Thank you!!